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Superseding
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MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT
S-348()/TSC-58

1. SCOPE

1.1 Scope.- This specification covers the transportable shelter facility part of the Terminal, Telegraph AN/TSC-58() which is capable of operation over three full-duplex or six half-duplex VF synchronous secure circuits; designated Shelter, Electrical Equipment S-348()/TSC-58. (See 6.5)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-6858	Welding, Aluminum, Magnesium, Non-Hardening Steels or Alloys and Titanium, Spot, Seam and Stitch
MIL-T-7928	Terminals Lug and Splice, Crimp Style, Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing of

FSC 5895

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STANDARDS

MILITARY

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment,Classification of Visual and Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single & Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ELECTRONICS COMMAND

SC-DL-564002	Shelter Electrical Equipment S-348()/TSC-58
SC-D-564107	Signal Wiring Diagram S-348/TSC-58
SC-F-578635	Power Wiring Diagram S-348/TSC-58

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the S-348()/TSC-58 for approval. (See 4.3)

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3.2 Construction.- Shelter,Electrical Equipment S-348()/TSC-58 shall be constructed in accordance with drawings as listed on SC-DL-564002. (See Para. 4.6)

3.3 Electrical installation.-

3.3.1 Power system wiring and components.- Installation of power wires and components, including: cable assemblies, receptacles, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly and wiring components constituting a three wire electrical power distribution system shall be installed and wired in accordance with the drawings. The completed in-place power distribution system shall meet the following requirements.

3.3.1.1 Continuity of conductors.- Electrical continuity shall exist between pin numbers 1,2 and 4 of Power Entrance Receptacle, U-238 and the respective circuit terminations thereof at receptacles, switches, lighting fixtures and lamp-holders when tested in accordance with 4.7.1 and 4.7.1.1.

NOTE: Test to verify the requirements of 3.3.1.1 (See 4.7.1 and 4.7.1.1) shall be accomplished following completion of power system wiring and prior to mounting of AC power receptacles in the wiremold duct.

3.3.1.2 Dielectric strength of insulation.- The insulation of all wires, cables and components shall withstand a potential of 600 V DC continuously applied for not less than 30 seconds when tested in accordance with 4.7.1 and 4.7.1.2.

3.3.1.3 Resistance of insulation.- The resistance of insulation between any two of the three circuit conductors, and; between each of the three circuit conductors and shelter ground shall be not less than 100 megohms when tested in accordance with 4.7.1 and 4.7.1.3.

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3.3.1.4 Power distribution and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as denoted by power system schematic Drawing SC-F-578635 when tested in accordance with 4.7.1.4 and the following:

a. Black Distribution Box 4.7.1.5 thru 4.7.1.13.

b. Red Distribution Box 4.7.1.14 thru 4.7.1.15.

3.3.2 Signal system wiring and components.- Installation of signal wires, cables and wiring components including binding posts, connectors, plugs, jacks, switches, terminal strips, and wiring devices, shall be installed and connected in accordance with the drawings. The completed in-place signal system installation shall meet the following requirements:

3.3.2.1 Dielectric strength of insulation.- The insulation of all wires, cables and components shall withstand a potential of 600 volts DC continuously applied for not less than 5 seconds when tested in accordance with 4.7.2.1.

3.3.2.2 Insulation resistance.- The resistance between each of the following shall not be less than 100 megohms when tested in accordance with 4.7.2.2:

a. Each wire and other remaining wires.

b. Each wire and shelter ground.

3.3.2.3 Continuity.- There shall be circuit continuity between the following wire terminations when tested in accordance with 4.7.2.3.

a. Each pin, 1A through 12B, of the 26-pair connector in the signal entrance box, and the corresponding termination at the Operating Positions, with respective CABLE/BINDING POST switch in CABLE position.

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b. Each pin, 13A through 24B, of the 26-pair connector and the corresponding termination on the patch panel, with respective CABLE/BINDING POST switch in CABLE position.

c. Each binding post, Outgoing Lines, in the signal entrance box and the corresponding termination at the Operating Positions, with respective CABLE/BINDING POST switch in BINDING POST position.

d. Each binding post, Spare Lines, in the signal entrance box and the corresponding termination on the patch panel, with respective CABLE/BINDING POST in the BINDING POST position.

e. Each binding post, SB-22 Lines, in the signal entrance box and the corresponding termination at the switchboard bracket.

f. Each binding post and each pin of the 26-pair connector, pairs 24 and 25,in the signal entrance box and the corresponding termination at the phone and intercom position.

3.3.3 Cable and cord assemblies.- Power cables,patch cables and phone cords shall be fabricated in accordance with the drawings and meet the requirements and tests of MIL-C-3885. (See 4.4)

3.3.4 Suppression of radio frequency interference.- The motors in the ventilating equipment used in the facility shall meet the emanation limits of MIL-STD-461 for tactical equipment. (See 4.4)

3.4 System operation.- With all operating components of Terminal Telegraph AN/TSC-58() installed and the equipment at each operating position connected in a closed loop, test message shall be sent and received by the respective send and receive equipments at the operating position. (See 4.7.3)

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3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage. (See 4.8)

3.6 Railroad transportation.- The S-348()/TSC-58 shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail. When so loaded, the shelter facility shall be subjected to and withstand the test of paragraph 4.9 without sustaining any buckling, delamination, or similar damage. Deformation to wall tile in the immediate area of the partitions and to the partitions shall not exceed 5/32 inch after completion of the rail tests.

3.7 Drops, flat and rotational.- The S-348()/TSC-58 shall withstand drops of 18 inches onto concrete. After all drop tests of 4.10.1 and 4.10.2 there shall be no buckling or delamination or similar damage. Deformation to partitions and adjacent wall area shall not exceed 5/32 inch after completion of all drops.

3.8 Watertightness.- The shelter facility shall be subjected to the test of 4.11. After the test there shall be no evidence of leakage into the facility.

3.9 Fording.- The shelter facility shall be capable of being immersed in water to a depth of 21 inches, measured from the bottom of the skids, for 1 hour without any leakage of water into the facility. (See 4.12)

3.10 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and when installed each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.13. There shall be no damage to the insert threads, the insert shall not have rotated from its original position, and there shall be no distortion of the shelter mounting surface or insert.

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3.11 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.12 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings.

3.13 Marking.-

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the equipment drawings.

3.13.2 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the center-of-gravity shall be determined by the contractor using the preproduction sample.

3.13.3 Nameplate.- Each S-348()/TSC-58 shall be serial numbered and provided with a nameplate in accordance with the drawings.

3.13.4 Interior and exterior marking.- All interior and exterior marking of the terminal shall be as depicted on the drawings.

3.14 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

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3.15 Seal, vapor.- All joints of the framing for the air-conditioner openings, the mail slot, and the power and signal entrance boxes shall be sealed to provide a vapor barrier against the entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retension of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be a one part sealant, silicone construction sealant SE-1204, as made by General Electric Corp., or equal.

3.16 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.17 Soldering.-

3.17.1 Solder and flux. Solder and flux shall be as defined on the drawings.

3.17.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

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3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of material.- Canvas,webbing,wood, etc shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use.(See 6.2e)

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3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.20 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

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3.20.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.4 and 4.14)

3.20.5 Identification of wiring.- Wiring shall be identified by colorcoding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.15) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.2 Construction
3.11 Cleaning
3.12 Finish
3.13 Marking
3.14 Riveting
3.15 Seal,vapor
3.16 Securing of parts
3.17 Soldering
3.18 Tropicalization
3.19 Welding
3.20 Wiring and cabling

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.17)

4.3 Preproduction inspection.- This inspection will be performed by the contractor on the first article unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (see Tables II,III, and IV, respectively). The preproduction inspection will normally be performed in this order:

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(a) Mounting inserts, (b) Power wiring, (c) Signal wiring, (d) System operation, (e) Vehicular transportation, (f) Rail transportation, (g) Flat drop, (h) Rotational drop, (i) Watertightness,(j) Fording. The last tests performed shall be a repeat of the power, signal, and system tests. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table I - Preproduction Inspection

Inspection (For additional preproduction inspection See 4.3)	Rqr Para	Insp Para
System operation	3.4	4.7.3
Vehicular transportation	3.5	4.8
Rail transportation	3.6	4.9
Drop,Flat	3.7	4.10.1
Drops,Rotational	3.7	4.10.2
Mounting inserts	3.10	4.13.1

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Cord assemblies	3.3.3
Suppression,RFI	3.3.4
Finish	3.12
Marking	3.13
Welding	3.19
Terminals	3.20.4

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4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (see 6.6) Test equipment for government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit of equipment shall be inspected for conformance to all the inspection and test requirements of Table II of this specification. Equipment conforming to the above requirement will be subject to verification utilizing the procedures for multi-level continuous sampling plan (CSP-M) of Standard MIL-STD-1235, unless otherwise specified

4.5.1.1 Group A sampling plans.- The Group A sampling plan for the AOQL's listed in Table II of this specification shall be selected from Tables VIII,IX and X of MIL-STD-1235 which are for f=½. The number of units in production interval shall be the entire contract quantity unless otherwise specified in the contract.

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Table II - Group A Inspection

Inspection	Rqr Para	Insp Para	AOQL Major	AOQL Minor
Visual and Mechanical	3.22	4.16	1.5(AQL 1.0%)	5(AQL 4.0%)
Electrical				**
Power Wiring				
Continuity	3.3.1.1	4.7.1.1	1%	
Dialectric strength	3.3.1.2	4.7.1.2	for	
Resistance of insulation	3.3.1.3	4.7.1.3		
Operational functions	3.3.1.4	4.7.1.4 thru 4.7.1.15	the entire group of	
Signal Wiring			power	
Dielectric strength	3.3.2.1	4.7.2.1		
Insulation resistance	3.3.2.2	4.7.2.2	and	
Continuity of conductors	3.3.2.3	4.7.2.3	signal wiring	

** All defects are major

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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III of this specification and to the procedures for the multi-level continuous sampling plan (CSP-M) of standard MIL-STD-1235, unless otherwise specified. Group B inspection shall normally be performed on units of production that have passed Group A inspection. The government reserves the right to require complete retest of any sample which is reworked after failing Group B inspection.

Table III Group B inspection

Inspection	Rqr Para	Insp Para	AOQL % Def (see 4.5.2.1)
Mounting insert	3.10	4.13.2	5 (AQL 4.0%)
Watertightness	3.8	4.11	3 (AQL 2.5%)
Fording	3.9	4.12	3 (AQL 2.5%)
Interchangeability	3.21	4.15	5 (AQL 4.0%)

4.5.2.1 Group B Sampling plans.- The Group B sampling plans for the AOQL's listed in Table III of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=½. The number of units in production interval shall be the entire contract quantity unless otherwise specified by the contract.

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

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Table IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Construction verification	3.2	4.6
Drops, flat and rotational	3.7	4.10
Solderless-terminal-lug connection	3.20.4	4.14

4.5.3.1 Sampling for inspection of drop tests.- One sample shall be selected at random from the first 50 units or fraction thereof, and one sample from each additional 150 units or fraction thereof, produced and subjected to the drop tests.

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Sampling for inspection of construction evaluation.- The equipment verification review in accordance with paragraph 4.6, shall be performed once on each part and assembly, randomly selected, and their installation into one of the first 10 shelter facilities produced.

4.5.3.4 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units or product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After

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corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.6 Equipment verification review.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings cited for construction (para 3.2). The EVR shall consist of the following:

a. An audit to establish that "as built" shelter facilities, including individual parts, their use in sub-assemblies and assemblies (components) and their final installation into the shelter forming the shelter facility, are in accordance with the cited end product drawings.

b. An audit to establish that "as built" configuration of shelter facilities, including parts, components and their installation meet the acceptance requirements (other than equipment specification performance) specified for each shelter part, components and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. The contractor shall compile an EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6a and 4.6b above.

(2) Discrepancies noted.

(3) Corrective action taken.

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d. The contractor shall submit to the government technical organization cited in the contract two (2) copies of his EVR reports no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government verification will be to the depth and extent necessaary to demonstrate that "as built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts and components of shelter facilities to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation and to re-assemble parts after inspection.

(2) As required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.7 Electrical installation tests.- The tests of 4.7.1 and 4.7.2 shall be performed on each Shelter Facility (S-348()/TSC-58) prior to installation of other major components comprising the AN/TSC-58 or dummy loads. Before applying power to the shelter facility, an electrical ground at the Power Entrance Box shall be made. Power to the shelter facility shall be 115 VAC, 60 cycles, single phase.

4.7.1 Power wiring, reference Drawing SC-F-578635.

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4.7.1.1 Continuity of conductors.- The shelter facility shall be tested for conformance to 3.3.1.1. Remove electrical loads as follows:

a. Fluorescent lamps and starters. DS-18 thru DS-33.

b. Incandescent lamps DS-34 thru DS-36.

c. Main power indicator lamp DS-1.

d. Power distribution box indicator lamps DS-2 thru DS-17.

e. Heaters, Nos 1 and 2.

f. Air-conditioners Nos 1 and 2.

g. In Air-conditioner relay box, wires to input side of terminal board TB-1 shall be transferred as follows: from terminal to terminal 3; from terminal 2 to terminal 5.

h. Disconnect leads to thermostat.

i. Disconnect leads to bell assembly E2 in bell box.

j. Disconnect leads to voltmeter and frequency meter.

k. Lamp in tape dryer.

NOTE. Position or insulate disconnected leads so that they do not short to each other or ground.

l. Electrically by-pass all filters in power entrance box and air-conditioner power box.

m. Remove and extend all connected power receptacles from wiremold so that outlet mounting brackets do not make electrical contact to the wiremold, except red power receptacles.

n. Place all circuit breakers and all switches in ON position. Place Blackout-Bypass switch in Bypass position.

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o. Using an aural or visual indicating device which produces a test current between 0.2 and 2.0 amperes AC when circuit continuity exists, measure continuity from Pin 1 of power entrance connector #1 in the power entrance box to the brass contact on each receptacle; the black wire termination at each lamp assembly; Pin 1 of connector #2 in entrance box.

p. Place all switches and circuit breakers, except main breakers, in OFF position.

q. Measure continuity from Pin 2 of power entrance connector #1 to the silver contacts on each receptacle; the white wire termination at each lamp assembly; Pin 2 of connector #2.

r. Place all circuit breakers and switches in OFF position.

s. Measure continuity from Pin 4 of power entrance connector #1 to each U-ground terminal on receptacles; shelter ground terminal in the power entrance box; Pin 4 of connector #2.

t. Replace and secure power receptacles.

4.7.1.2 Dielectric strength of insulation.-

a. With all loads removed as in 4.7.1.1, place all circuit breakers and switches in ON position.

b. Employing method 301 of MIL-STD-202, connect a DC test instrument connected between Pins 1 and 2 of power entrance connector #1. Adjust for a value of 600 volts and maintain for period of 30 seconds.

c. Repeat 4.7.1.2b with test instrument connected between Pins 1 and 4 of connector #1.

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d. Repeat 4.7.1.2b with test instrument connected between Pins 2 and 4 of connector #1.

NOTE: To eliminate danger of shock. Discharge conductors after each test by connecting a 10-60 ohm wire wound resistor or equivalent test lamp to the pins tested.

4.7.1.3 Resistance of insulation.-

a. With all loads removed as in 4.7.1.1, place all circuit breakers and switches in ON position.

b. Employing method 302 of MIL-STD-202, connect test instrument between Pins 1 and 2 of power entrance connector #1. Adjust for a value of not less than 100 nor more than 600 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance.

c. Repeat 4.7.1.3b with test instrument connected between Pins 1 and 4 of connector #1.

d. Repeat 4.7.1.3b with test instrument connected between Pins 2 and 4 of connector #1.

NOTE: To eliminate danger of shock. Discharge conductors after each test by connecting a 10-60 ohm wire wound resistor or equivalent test lamp to the pins tested.

4.7.1.4 Operational functions tests.-

a. Replace and reconnect all electrical loads and remove filter by-passes cited in 4.7.1.1.

b. Place all circuit breakers and switches in OFF position.

c. Insert power line plugs of heaters, air-conditioners, and tape over into their respective receptacles.

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d. Place Blackout-By-pass switch in By-pass or ON position.

e. Place a 115 volt AC test lamp with U-grounding type plug into each receptacle (in power entrance box, and inside shelter).

f. Apply 115 volt AC (nominal), 60 cycles, power to power connector #1. Test lamps in receptacles in power entrance box (this outlet protected by a circuit breaker with reset button in power entrance box; access from inside shelter) shall light; neon lamp DS1 shall light; voltmeter on Black Distribution Box and frequency meter on Red Distribution Box shall indicate nominal 115 V AC and 60 cycles respectively.

4.7.1.5 Black power distribution, Circuit No.1.-

a. Place main circuit breaker CB18 in ON position.

b. Place CB2 (lights) in ON position, neon indicator shall light.

c. Place fluorescent light switch S1 in ON position. Inside bank of lamps shall light. Place fluorescent light switch S2 in ON position. Outside bank of lamps shall light. Place switches S1 and S2 in OFF position.

d. Place incandescent light switch S5 (at desk) in ON position; incandescent lamps over desk shall light. Return switches S1 and S2 to ON position.

e. Place circuit breaker CB2 in OFF position; all lamps shall extinguish.

f. Place circuit breaker CB2 in ON position and close shelter door; lamps remain lighted. Place Blackout-By-pass switch S4 to BLACKOUT position; lamps remain lighted. Open shelter door; lamps shall extinguish. Place switch S4 in BY-PASS position; lamps shall light.

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4.7.1.6 Black power distribution,Circuit No.2.-

a. Place circuit breaker CB3 in ON position. Neon indicator shall light.

b. Turn Air-Conditioner #1 Fresh Air Switch to OPEN position and Fan-Cool switch to FAN. Air current shall emit from air-conditioner. Record current indicated on panel meter.

c. Turn Fan-Cool switch to COOL position and Fresh Air switch to CLOSED position. Turn thermostatic control at desk to lowest temperature. After a period of about 5 minutes the temperature of the air from the air-conditioner shall be cooler than the ambient air in the shelter. The unit shall function smoothly and the ammeter reading shall have increased above the first observed reading.

d. Place circuit breaker CB3 in OFF position; air-conditioner shall stop functioning. Return Fan-Cool switch to OFF position.

4.7.1.7 Black power distribution,Circuit No.3.-

a. Place circuit breaker CB4 in ON position.

b. Repeat 4.7.1.6c and d for Air Conditioner #2.

c. Place circuit breaker CB4 in OFF position; air-conditioner shall stop functioning. Return Fan-Cool switch to OFF position.

4.7.1.8 Test of Time Delay Box.-

a. Turn thermostat control for air-conditioners at desk to highest temperature setting. Place circuit breakers CB3 and CB4 in ON position. Place Fan-Cool switches in COOL position.

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b. Turn thermostat control to lowest temperature setting. Air-Conditioner #1 shall begin to function followed, after a 10 second minimum delay, by Air-Conditioner #2.

c. Turn off air-conditioner and breakers CB3 and CB4.

4.7.1.9 Black power distribution,Circuit No.4.-

a. Place circuit breaker CB5 in ON position (heater #2). Neon indicator shall light.

b. With the heater switch set at FAN position the heater shall blow air. After a few minutes at the HEAT position (with thermostat turned up) the heater shall blow warm air, and shut off with the switch in the straight up position. Place heater in HEAT position and turn thermostat setting to colder temperature setting. When the lower temperature setting is reached, a click shall occur and the panel ammeter shall drop about 10 amps when the heater shuts off. The fan shall continue to blow.

c. With the heater in one of its operating positions, turn off breaker CB5. The heater shall stop. Return heater switch to straight up (OFF) position.

4.7.1.10 Black power distribution,Circuit No.5.-

a. Place circuit breaker CB6 in ON position. Neon indicator shall light.

b. Repeat 4.7.1.9b and c for heater #1.

4.7.1.11 Black power distribution,Circuit No.6.-

a. Place circuit breaker CB7 in ON position.

b. The 115 VAC test lamp in each of the eight TH-22()/FG outlets (J11 thru J14) shall light.

c. Place breaker CB7 in OFF position.Test lamps shall extinguish.

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4.7.1.12 Black power distribution,Circuit No.7.-

a. Place circuit breaker CB8 in ON position.Neon indicator shall light.

b. The 115 VAC test lamp in intercom outlet at the desk shall light.

c. Place phone switch at desk to ON position. Red phone lamp shall light. Place phone switch OFF.

d. Repeat c above for Intercom switch and lamp.

e. Push Entrance button in Power Entrance Box. Bell shall ring.

f. Place tape oven switch ON. Indicator lamp and lamp in oven shall light.

g. Place breaker CB8 in OFF position. Lamps and bell shall not function.

4.7.1.13 Black power distribution,spare circuits.-

a. Place circuit breaker CB9 in ON position. Neon indicator shall light. Place breaker CB9 in OFF position.

b. Repeat 4.7.1.13a for breaker CB10 and lamp DS10; breaker CB11 and lamp DS11.

4.7.1.14 Red power distribution,Circuit No.1.-

a. Place main circuit breaker CB19 in ON position.

b. Place circuit breaker CB12 in ON position. Neon indicator shall light.

c. The 115 VAC test lamp in each of the two position 1 outlets (J3) shall light.

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d. The test lamp in AUX 1 outlet shall light.

e. Place breaker CB12 in OFF position. The lamps shall extinguish.

4.7.1.15 Red power distribution,Circuit Nos 2 thru 6.- Repeat test of 4.7.1.14 for circuit breakers CB13 thru CB17 with test lamps in POSITION 2 thru 6 outlets and AUX 2 thru 6 outlets respectively.

4.7.2 Signal wiring.- Reference Drawing SC-D-564107.

4.7.2.1 Dielectric strength of insulation.- Dielectric strength of insulation tests which verify the requirements of 3.3.2.1 shall be made on each signal circuit indicated by the following paragraphs,employing method 301 of MIL-STD-202, using a non-destructive dielectric test instrument and suitable test jigs or automatic tester.

a. Place all CABLE/BINDING POST signal switches S1 thru S12 in CABLE position.

b. Place switch on the DUMMY BOX-AUX switch box at each of the six Operating Positions in the DUMMY BOX position.

c. Place switches on the TELEPHONE-INTERCOM BOX in the ON position.

d. Connect following to the shelter ground terminal:

(1) All pins of the 26-pair connector in the signal entrance box.

(2) All binding posts in the signal entrance box except 26A and 26B, and 25A and 25B.

e. Insert Phone Plug,Black,Send,at TH-22/TG No.1 rack position into Position #1,Line 1,MON Jack on the Patch Panel.

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f. Insert Phone Plug Red,Rec,at TH-22/TG No.1 rack position into Position #1, Line 2, MON Jack on the Patch Panel.

g. Repeat e and f for remaining plugs at TH-22/TG Nos 2 thru 6 rack positions and the corresponding Line Mon Jacks for Positions 2 thru 6 on the Patch Panel.

h. Insert Phone Plug,Black,Send at TH-22/TG Spare No 1 rack position into Spare Line 13, MON Jack on the Patch Panel.

i. Insert Phone Plug,Red,Rec,at TH-22/TG Spare No.1 rack position into Spare Line 14, MON Jack on the Patch Panel.

j. Repeat h and i for plugs at TH-22/TG Spare No.2 rack position and corresponding MON Jacks,Spare Lines 15 and 16 on the Patch Panel.

k. Using a patch cord whose ends terminate in black phone plugs, insert one phone plug into Spare #1 TH-22/TG, Line,4WS Jack on the Patch Panel; the other plug into Spare Line 17,MON Jack.

l. Using a patch cord whose ends terminate in red phone plugs, insert one plug into Spare #1 TH-22/TG Line,2W-4WR Jack on the Patch Panel; the other plug into Phone Jack,Spare Line 18,MON Jack.

m. Repeat k and l for Spare #2 TH-22/TG Line Jacks and MON Jacks,Spare Lines 19 and 20.

n. Using a patch cord whose ends terminate in red phone plugs,insert one plug into Phone Jack SB-22/PT, OPER on the Patch Panel; the other plug into Spare Line 24, MON Jack.

o. At Operating Positions 1 thru 6,top shelf,connect leads Nos 1 and 2 and connect leads Nos 4 and 5;lower shelf connect leads Nos 2 and 5. Position or insulate all other leads at the Operating Positions at the desk (phone,intercom,switch-boards) so they do not short to each other or to ground.

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p. Disconnect ground connection from Pin 1A of 26-pair connector. Connect tester between Pin 1A of 26-Pair connector and shelter ground terminal. Adjust tester to 600 VDC output and maintain this potential for a continuous period of not less than 5 seconds.

q. Disconnect Pin 1A of 26-pair connector from tester and reconnect Pin 1A to ground terminal.

r. Repeat p and q for remaining pins 1B thru 26B of 26-pair connector.

s. Place all CABLE/BINDING POST switches in BINDING POST position.

t. Repeat procedure of p and q for all Binding Posts in the signal entrance box.

NOTE. To eliminate danger of shock. Discharge conductors following each test by connecting a 500 ohm wirewound resistor or equivalent test lamp to the pins tested.

4.7.2.2 Insulation resistance.- Employing method 302 of MIL-STD-202, using a megohmmeter or equivalent tester, and suitable test jig or automatic tester, insulation resistance shall be measured. Repeat 4.7.2.1a thru 4.7.2.1t except in 4.7.2.1p, replace 600 VDC with 500 VDC. Upon completion, remove all plugs from patch panel, connections at Operating Positions, and grounds in entrance box.

NOTE. To eliminate danger of shock. Discharge conductors following each test by connecting 500 ohm wire wound resistor or equivalent test lamp to the pins tested.

4.7.2.3 Continuity of conductors.- Continuity of signal circuits shall be tested using a low resistance continuity tester which produces a test current of less than 0.010 ampere.

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a. Place signal switches in CABLE position.

b. At the TH-22/TG No.1 rack position, connect the White lead of the SEND wire pair (line side) to the tip of the BLACK send plug (loop side) and the BLUE lead of SEND wire pair to the sleeve of the send plug.

c. At the TH-22/TG No.1 Rack Position,connect the Red lead of the Rec.Wire pair(line side) to the tip of the Red Receive plug(loop side) and the Black lead of the Receive wire pair to the sleeve of the plug.

d. Measure continuity from Pin 1A in entrance box and lead No.5, lower shelf of Position #1; from Pin 1B and lead No.7, upper shelf,Position #1; from Pin 2A to lead No.4, lower shelf,Position #1; from Pin 2B to lead No.5, upper shelf,Position #1.

e. Connect leads Nos 1 and 2, upper shelf,Position #1; leads 4 and 5,upper shelf,Position #1;leads 2 and 5, lower shelf,Position #1.

f. Measure continuity from Pin 1A to Lead No6, upper shelf,Position #1; from Pin 2B to lead No.3,lower shelf.

g. Insert one plug of a patch cord into Equipment Send Jack,Position No.1. There shall be no continuity from Pin 2A to the tip of the other plug or from Pin 2B to sleeve. Repeat with Patch Cord plug in Position No.1 TH-22/TG Line,4WS Jack.

h. Insert one plug of a Patch Cord in Position No.1, TH-22/TG Loop,send Jack and measure continuity from Pin 1A to the tip of the other plug and Pin 1B to sleeve. Repeat with patch cord plug in Position No.1, Line 1,4WS Jack;Position No.1,Line 1,MON Jack.

i. Insert one plug of a patch cord into Position No.1,Equip Rec Jack. There shall be no continuity from Pin 2A to the tip of the other plug and from Pin 2B to sleeve. Repeat with patch cord plug in Position No.1,TH-22/TG Line,2W4WR Jack.

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j. Insert one plug of a patch cord in Position No.1, TH-22/TG loop Rec Jack and measure continuity from Pin 2A to the tip of the other plug and Pin 2B to sleeve.Repeat with patch cord in Position No.1, Line 2, 2W4WR Jack;Position No.1 Line 2,MON Jack.

k. Remove all plugs from Patch Panel and connections at Position #1.

l. Measure continuity from Pin 1A to lead White of 4WS pair at TH-22/TG No.1 rack position; Pin 1B to lead Blue of 4WS pair; Pin 2A to lead Red of 2W4WR pair; Pin 2B to lead Black of 2W4WR pair.

m. At Interconnecting Box,Position 1, measure continuity from Binding Post E1 to Connector J8,Pin C;E2 to J7,Pin B;E3 to Pin A of J3, J4, J7 and J8; E4 to J3, Pin B;E5 to J4,Pin C;J7,Pin C to J8, Pin B;J4,Pin B to J3,Pin C.

n. At Dummy Box-Aux switch box,Position 1,with switch in Dummy Box position,measure continuity from J1, Pin B to J3,Pin B;J1,Pin C to J2, Pin B;J2,Pin C to J3, Pin C;J1,Pin A to J2,Pin A to J3,Pin A. With the switch in the AUX position,using an ohmmeter,measure 27,000 ohms ± 5% from J1,Pin B to J3,Pin B.

o. At both upper and lower shelf of Position 1, measure continuity from ground wire termination to shelter ground terminal in the power entrance box; at Interconnecting Box,Position 1,measure continuity from plug E3 to shelter ground terminal;measure continuity from TH-22/TG 1 case to shelter ground terminal.

p. Repeat 4.7.2.3a thru o for Positon #2 and Pins 3A,3B,4A and 4B;Position #3 and Pins 5A,5B,6A, and 6B; Position #4 and Pins 7A,7B,8A, and 8B;Position #5 and Pins 9A,9B,10A, and 10B; Position #6 and Pins 11A,11B,12A and 12B.

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q. Insert one plug of a patch cord into Spare No.1 TH-22/TG Loop, Send Jack. Measure continuity from tip of other plug to tip of Spare No.1 TH-22/TG rack position Send Plug; between sleeves of the plugs.

r. Repeat q above with patch cord in Spare No.1 TH-22/TG Loop,Rec Jack; Spare No.1 TH-22/TG Line,4WS, Jack; Spare No.1 TH-22/TG Line 2W4WR Jack. Measure continuity to corresponding Spare No.1 rack position Rec plug;leads of line 4WS pair; leads of Line 2W4WR pair.

s. Repeat q and r above for Spare No.2 TH-22/TG.

t. Insert one plug of a patch cord into Spare Line 13,4WS Jack. Measure continuity from Pin 13A in entrance box to tip of other plug; Pin 13B to sleeve. Repeat with patch cord in Spare Line 13 MON Jack.

u. Repeat t above Spare Lines 14 thru 24.

v. Place all signal switches in BINDING POST position. Repeat t and u above for all Line jacks and Binding Post pairs 1 thru 24.

w. Measure continuity from SB-22/PT,Pos 1 Binding Post A to corresponding lead at the desk. Repeat for remaining lines 1B thru 12B.

x. At desk,connect wires of Phone pair together; place Phone switch in ON position; in entrance box measure continuity from Pins 25A and 25B and Binding Posts 25A and 25B. Disconnect wires at desk and place phone switch in OFF position.

y. Connect a Telephone Set TA-312/PT to Binding Posts 25A and 25B. Crank the TA-312/PT hand-crank. The clacker in the Phone-Interome Switchbox shall sound. Disconnect the phone.

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z. At desk,connect wires of Intercom pair together; place Intercom switch in ON position; in entrance box,measure continuity from Pins 26A and 26B and Binding Posts 26A and 26B. Disconnect wires at desk and place Intercom switch in OFF position.

aa. Insert one plug of a patch cord into SB-22/PT OPER Jack on Patch Panel. Measure continuity from tip and sleeve of other plug to red and orange leads respectively for OPER at desk.

ab. Disconnect all plugs from Patch Panel.

4.7.3 System operation.-

4.7.3.1 Position 1 Send,Position 2 Receive.-

a. At all positions verify that the TD Send,TR Send, and Rec wire pairs are connected to their respective terminals on the TT-76()/GGC and that the KYBD and PTR wire pairs are connected to their respective terminals on the TT-98()/FG. Connect all line pairs, 1 thru 12, and operator pair to respective binding posts on SB-22()/PT. Install batteries in switchboard. The Send and Rec plugs and wire pairs shall be connected to their respective jacks and terminals on the TH-22()/TG. Plug in all power cords to their respective outlets. Turn on all circuit breakers. Install tape roll in TT-76()/GGC and page roll in TT-98()/FG.

b. At the Patch Panel, using a patch cord,Jumper Position #1, Line 1, MON to Position #2,Line 4,MON.

c. Turn on equipments associated, with Positions 1 and 2.

d. Using the keyboard of the Position 1 TT-76()/GGC prepare a test tape containing a message utilizing all letters. The message shall be repeated twice. (Selector switch in position).

e. Using the keyboard of the Position 1 TT-98()/FG repeat the test message two additional times.

f. Using the transmitter-distributor on the Position 1 TT-76()/GGC duplicate the message six additional times, completing the test tape.

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g. The ten messages on the test tape shall be received simultaneously at Position 2 on the TT-76()/GGC tape and on TT-98()/FG page copy.

4.7.3.2 Tests of remaining Send and Receive circuits.-

a. Position 2 Send - Position 3 Receive; Position 3 Send - Position 4 Receive; Position 4 Send - Position 5 Receive; Position 5 Send - Position 6 Receive; Position 6 Send - Position 1 Receive shall be tested by repeating the test procedure of 4.7.3.1b through g using the associated jacks and teletype equipments.

b. Disconnect all jumpers.

4.7.3.3 Telephone circuit operation.-

a. Connect Telephone Set TA-312()/PT to Phone wire pair at desk and put phone switch to phone position. Red phone lamp shall light.

b. Connect a second phone to binding post pair 25A and 25B and "Ring-up" desk phone. Determine that conversation is established.

c. Disconnect second phone and put phone switch in OFF position.

4.7.3.4 Intercom circuit operation.-

a. Connect Intercommunication Station LS-147()/FI to intercom wire pair at desk and put intercom switch to intercom position. Red intercom lamp shall light.

b. Connect a second intercom to binding post pair 26A and 26B and call into desk intercom. Determine that conversation is established.

c. Disconnect second intercom and put intercom switch in OFF position.

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4.7.3.5 SB-22()/PT circuit operation.-

a. Using a patch cord, patch the SB-22/PT OPER Jack to Position #6, TH-22/TG Line, 2W4WR Jack.

b. Outside the shelter, in the signal entrance box, jumper from Position #1,Line 2A (2W-4WR) Binding Post to SB-22/PT Binding Post 2A; jumper from Position #2, Line 2B (2W-4WR) Binding Post to SB-22/PT Binding Post 2B. Similarly jumper Position #2, Line 4A and 4B (2W-4WR) to SB-22/PT 4A and 4B.
NOTE: Operating Positions and switchboard lines are arbitrary selections.

c. Set TH-22/TG units 1,2, and 6 for 2 wire operation and place circuit breakers and power switches associated with the equipments of these Positions to ON.

d. Originate ring with TH-22/TG #1. Indicator at line 2 of switchboard shall operate. Insert Operator Plug in calling party jack. Establish teletype communication using equipments at Positions 1 and 6. Remove Operator Plug from calling party jack and insert it into switchboard line 4. Ring with TH-22/TG #6. Establish teletype communication using equipments at Positions 4 and 6.

e. Insert plug of called party (line 4) into jack of calling party (line 2). Establish teletype communication using equipments at Positions 1 and 2, and monitor transmission at Position 6.

f. Remove the Operator Plug from called line and insert in Operators Jack.

g. There shall be no change in communication between Positions 1 and 2.

4.8 Vehicular transportation test.- The facility shall be secured to a government furnished 2-½ ton, M-35,truck and transported over the specially prepared course described in 4.8.1 for 10 laps at a speed of 5 mph, 10 laps at a speed of 10 mph,10 laps at a speed of 15 mph,and 10 laps at a speed of 18-20 mph.A lap is defined as traversing the course in one

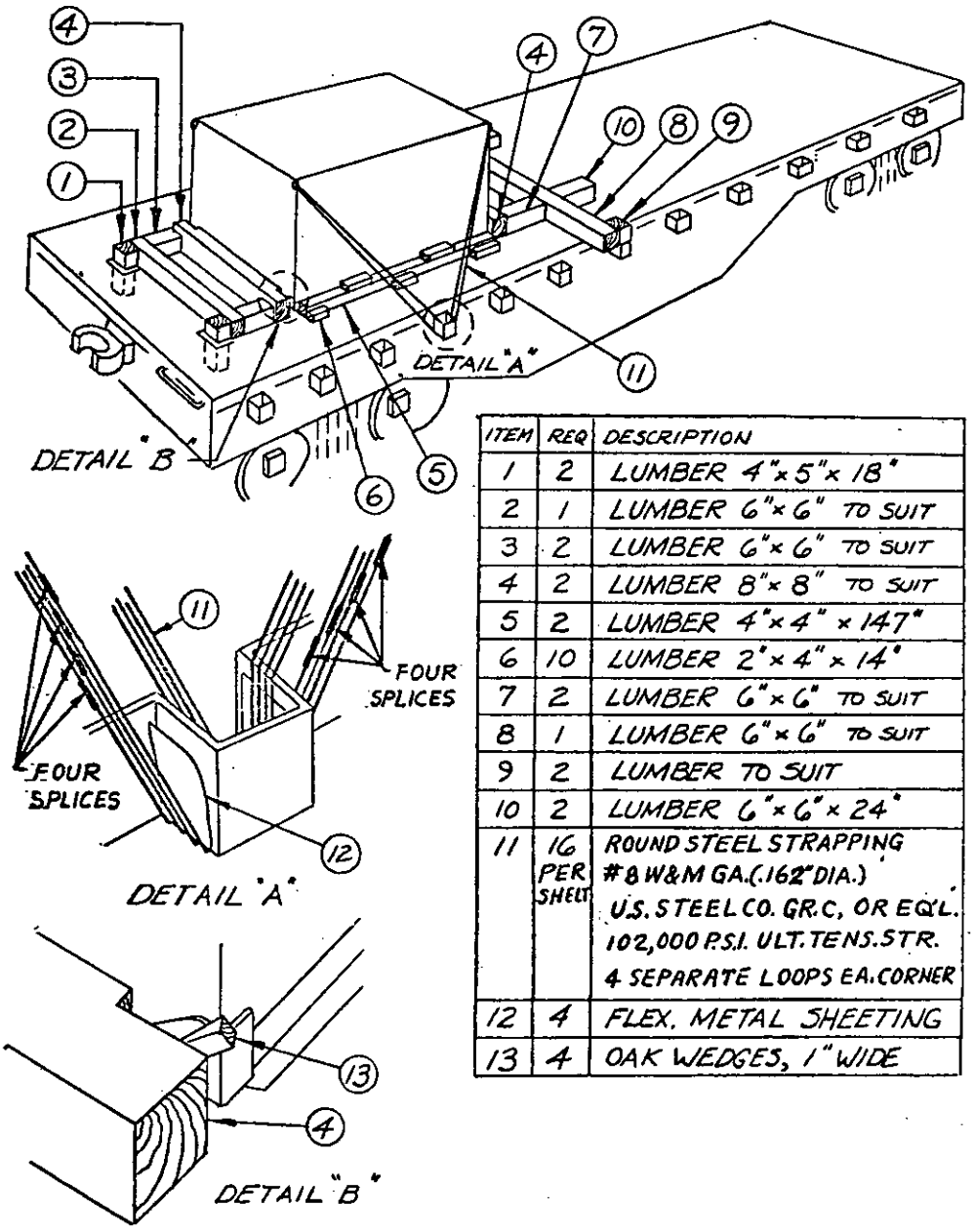
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direction. The facility shall meet the requirements of paragraph 3.5 after completion of the test. (NOTE: The facility shall be loaded to simulate the AN/TSC-58() using live and dummy loads (See 6.2(b) for this test).

4.8.1 Vehicular transportation course. The course for the military truck transportation shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1 inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the width of the vehicle and anchored securely by any convenient means. The surface on which the timbers are placed shall be hard and level.

4.9 Rail transport impact test.- The shelter facility, loaded as specified in paragraph 4.8 secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a weighted gondola car. The test shall be performed on a level stretch of track. The test flatcar with its timber blocking and shelter facility tiedown strapping shall be as specified in Figure 1 and shall meet the provisions of the test as specified in paragraph 4.9.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The weighted gondola car shall meet the provisions of the test as specified in paragraph 4.9.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the Governments Technical Representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars all with brakes off. This test shall consit of four impacts. Two impacts shall be performed with the door end of the shelter facing the loaded gondola; and two impacts shall be performed with the door end facing in the opposite direction from the loaded gondola. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"×5"×18"
2	1	LUMBER 6"×6" TO SUIT
3	2	LUMBER 6"×6" TO SUIT
4	2	LUMBER 8"×8" TO SUIT
5	2	LUMBER 4"×4"×147"
6	10	LUMBER 2"×4"×14"
7	2	LUMBER 6"×6" TO SUIT
8	1	LUMBER 6"×6" TO SUIT
9	2	LUMBER TO SUIT
10	2	LUMBER 6"×6"×24"
11	16 PER SHEET	ROUND STEEL STRAPPING #8 W&M GA.(.162" DIA.) U.S. STEEL CO. GR.C, OR EQ'L. 102,000 P.S.I. ULT. TENS. STR. 4 SEPARATE LOOPS EA. CORNER
12	4	FLEX. METAL SHEETING
13	4	OAK WEDGES, 1" WIDE

FIGURE 1 - SHELTER SECURED ON FLATCAR

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is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

4.9.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 and 8 x 8 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The 8 x 8 inch blocking shall be notched to provide 1/4 inch clearance around towing eye castings of the shelter. A wedge shall be installed into this clearance as shown in the figure referred to above, at each of the four corners of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred above. All blocking around the shelter facility shall fit tightly against the shelter. All other blocking shall be firmly secured as shown in the figure. The tie down strapping shall be installed using a strapping machine (US Steel Corp. Model G.T. or equal). If the tie down strapping loosens or breaks during any run, it shall be replaced before the next run with new wire and restrapped as above.

4.9.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete, metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.10 Drop tests.-

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4.10.1 Flat drop.- The S-348()/TSC-58 shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the assemblage shall meet the requirements of 3.7. NOTE: The facility shall be loaded as in 4.8.

4.10.2 Rotational drop.- The S-348()/TSC-58 shall have one bottom edge placed on a nominal 4 inch high timber. The opposite edge of the terminal shall be raised 18 inches and allowed to fall freely onto a hard concrete surface. This impact shall be performed on each bottom edge of the terminal. Upon completion of these drops the assemblage shall meet the requirements of 3.7. NOTE: The facility shall be loaded as in 4.8.

(In each drop, the 4 inch high timber shall be placed under the skids and the 18 inch measurement shall be taken from the bottom of the skids.

4.11 Watertightness test.- The S-348()/TSC-58, without the use of any external sealing, caulking,taping etc shall be subjected to the following two part test (NOTE: Perform tests with inside cover of entrance box removed):

a. Each of the five exposed surfaces of the shelter, with covers closed,air-conditioners installed shall be simultaneously sprayed with water from nozzles,model G 29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois,or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches and the row of nozzles in line with the air-conditioners where the 19 inches shall be measured from the extreme vertical surface of the air-conditioner.

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(See Figure 2 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.11, the bottom row of nozzles can be eliminated with the exception of the nozzles facing the door-end panel). Upon completion of the above test, the facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

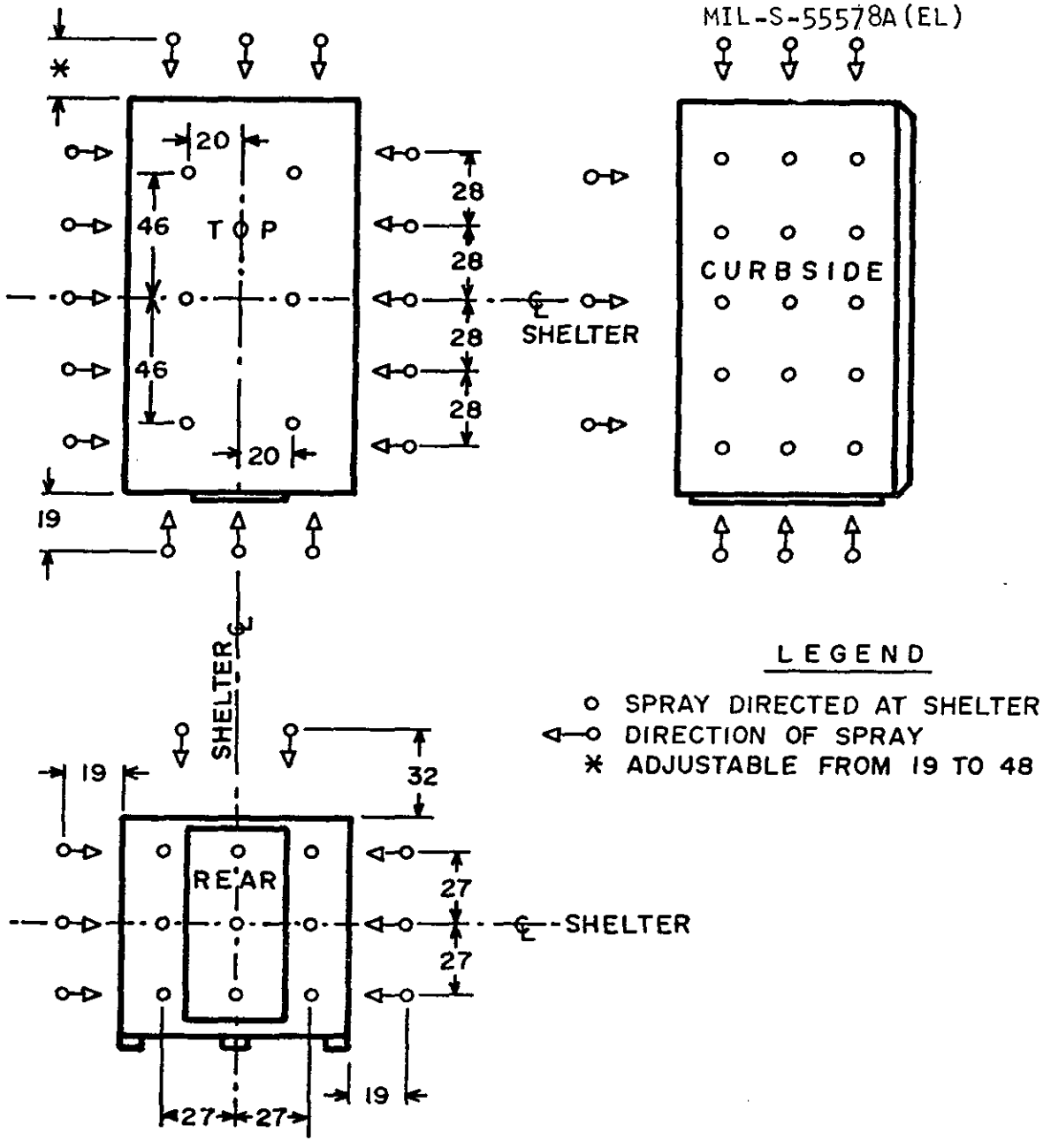
b. Repeat the test of (a) with covers open to their normal operating position, one air-conditioner removed and replaced by its opening cover. The nozzle in line with the opening cover shall be 19 inches away from that cover.

4.12 Fording test.- The S-348()/TSC-58 shall be immersed in water to a depth of 21 inches and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.13 Mounting insert tests.-

4.13.1 Preproduction insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter. This test shall be performed prior to the installation on any brackets, racks, etc.

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LEGEND

- o SPRAY DIRECTED AT SHELTER
- ◁—o DIRECTION OF SPRAY
- * ADJUSTABLE FROM 19 TO 48

NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 2
PAGE 41

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4.13.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample facility. This test shall be performed prior to the installation of any brackets, racks,etc.

Table V - Mounting Insert Loads

Insert size	Preproduction Test		Production Test	
	Axial(lbs)	Torque(in-lbs)	Axial (lbs)	Torque(in-lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60

NOTE: Torque loads applied through bolt whose head holds a stationary plate against head of insert.

4.14 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.15 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

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b. Size and mounting dimensions of racks and cabinets.

c. Size and mounting dimensions of equipment drawers.

d. Size and mounting dimensions of air-conditioner opening covers.

4.16 Visual and mechanical inspection.- The equipment shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VI.

Table VI-Classification of Visual & Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated.
	2. Aircraft loading data plate,instruction or nameplate omitted,incorrect or illegible.
	3. Welds - cracked or porous.
	4. Sealer improperly applied (holes, separations, or lack of adhesion).
	5. Mounting inserts missing or mislocated.
	6. Burrs or sharp edges in wire duct not removed.
	7. Equipments or bracketry loose or missing.
	8. Drawers,slides,doors,etc inoperative.
	9. Power receptacles,switches,jacks, wire ducts or other electrical components loose,improperly located or inoperative.
	10. Locking or holding devices missing or inoperative.
	11. Gaps between sections of wire duct exceed 1/16".
Minor	1. Drawers,slides,door,not easily operable.
	2. Fastening devices difficult to operate.
	3. Finish - abrasions or scratches.

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4.17 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.18 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7.1 and 4.7.2	Power and Signal wiring
Paragraph 4.16	Visual and Mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the S-348()/TSC-58 shall be in accordance with applicable requirements of Specification MIL-S-55505.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-348()/TSC-58 is the shelter facility portion of the AN/TSC-58() and it is used to house the following operational equipments in modified S-280(B)/G Shelter:

6 ea TT-76()/GGC Teletype
6 ea TT-98()/FG Teletype
8 ea TH-22()/TG Telegraph Terminal
1 ea SB-22()/PT Switchboard
8 ea TT-523()/GGC Keyers
6 ea Security Equipment (optional)
1 ea TA-312()/PT Telephone
1 ea LS-147()/FI Intercom

forming the telegraph terminal.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title,number, and date of this specification and any amendment thereto.

(b) Preproduction and Group C inspection of the S-348()/TSC-58 requires use of contractor fabricated dummy loads for the communications security equipments (6 ea) and government furnished live loads (teletype and telephone components) unless otherwise specified in the bid request and contract.

(c) A list of government furnished equipments required as part of the S-348()/TSC-58 on order.

(d) A list of equipments to be government loaned for testing purposes.

(e) Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government,pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3)

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6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: AN/TSC-58R. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
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Preparing Activity:
Army - EL

Project No. 5895-A112

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-55578A(EL), SHELTER, ELECTRICAL EQUIPMENT S-348( )/TSC-58

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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